

武汉大学数学与统计学院 2024-2025 第一学期

抽象代数 期末 ZUOKANG

Throughout this exam, all rings are assumed to have unity 1.

1. Let $\theta \in \mathbb{C}$ be a solution of $x^4 - 15x + 5 = 0$. (10 points)

(a) Compute $[\mathbb{Q}(\theta) : \mathbb{Q}]$. (4 points)

(b) Find a polynomial $g \in \mathbb{Q}[x]$ such that $\frac{1}{1 + \theta + \theta^2} = g(\theta)$. (6 points)

2. Let R be an integral domain, and let

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in R[x].$$

Prove that $f(x) \in U(R[x])$ (i.e., $f(x)$ is a unit in $R[x]$) if and only if $a_0 \in U(R)$ and $a_1, a_2, \dots, a_n$ are nilpotent elements of R . (10 points)

3. 忘记了, 但是不难。青春有遗憾才完整。 (8 points)

4. Let $K \supseteq L \supseteq F$ be field extensions. Consider $\alpha \in K$, which is algebraic over F . If the minimal polynomial of α over F is $f(x) \in F[x]$ and $f(x)$ is irreducible over L , show that $F(\alpha) \cap L = F$. (10 points)

5. Let $E \supseteq F$ be a finite field extension and $L = F(u_1, \dots, u_n)$, $K = F(v_1, \dots, v_m)$ be two intermediate fields. (10 points)

(a) Let LK be the smallest subfield of E with $L \subset LK$ and $K \subset LK$. Show that LK is equal to $F(u_1, \dots, u_n, v_1, \dots, v_m)$. (4 points)

(b) If L is a finite Galois extension of $L \cap K$, show that LK is also a finite Galois extension over K . (6 points)

6. Suppose that $\text{char}(F) = 0$, and let field $K := F(t)$, where t is transcendental over F . Let $L_1 = F(t^2)$ and $L_2 = F(t^2 + t)$, which are subfields of K . (12 points)

(a) Compute $[K : L_1]$ and $[K : L_2]$. (4 points)

(b) Determine $\text{Gal}(K/L_1)$ and $\text{Gal}(K/L_2)$. (4 points)

(c) Show that $L_1 \cap L_2 = F$. (4 points)

7. Let $f(x) \in \mathbb{Z}[x]$ be an irreducible polynomial of degree four with splitting field E and $\text{Gal}(E/\mathbb{Q}) = S_4$. Let α be a root of $f(x)$ and set $K = \mathbb{Q}(\alpha)$. (15 points)

(a) Show that over K , one has $f(x) = (x - \alpha)g(x)$ for some irreducible polynomial $g \in K[x]$. (3 points)

(b) Show that $\text{Gal}(E/K) \cong S_3$. (3 points)

(c) Show that there is no field F such that $K \supseteq F \supsetneq \mathbb{Q}$. (4 points)

Hint: If there is such an F , note that $|\text{Gal}(E/F)| = 12$

(d) Is the extension $K/\mathbb{Q}$ Galois? Give your reason. (2 points)

(e) Let $\alpha_1 = \alpha$ and $\alpha_2, \alpha_3, \alpha_4$ be four roots of f over its splitting field. And let $\beta := \prod_{1 \leq i < j \leq 4} (\alpha_i - \alpha_j)$. Show that $[\mathbb{Q}(\beta) : \mathbb{Q}] = 2$. (3 points)

8. Let $E = \mathbb{Q}(\sqrt[3]{4} - \sqrt[3]{2}, i)$ and $F = \mathbb{Q}(i)$. (11 points)

(a) Compute $[E : \mathbb{Q}]$. (3 points)

(b) Is the extension E/F Galois? Give your reason. (3 points)

(c) Is the extension $E/\mathbb{Q}$ Galois? Give your reason. (3 points)

(d) Compute the group $\text{Gal}(E/F)$. (2 points)

9. Let $a = \sqrt[4]{2}$, and let $E = \mathbb{Q}(a, i)$. (14 points)

(a) Compute $[E : \mathbb{Q}(\sqrt{-2})]$. (2 points)

(b) Is the extension $E/\mathbb{Q}(\sqrt{-2})$ Galois? Give your reason. (2 points)

(c) Compute the group $\text{Gal}(E/\mathbb{Q}(\sqrt{-2}))$. (3 points)

(d) Find all intermediate fields K of $E/\mathbb{Q}(\sqrt{-2})$ and determine the group $\text{Gal}(E/K)$ for each K . (5 points)

(e) Are these $K/\mathbb{Q}(\sqrt{-2})$ Galois? Give your reason. (2 points)